

ARRAY

Array:-

- Array is pre defined class present in java. Lang package. Array is a non primitive data type in java .array is continues block of memory which is used to store multiple value.

To store or access the value in an array we need two things

- 1) Array reference variable
- 2) Index.

- Array stores homogenous value.

Characteristics of array:-

- 1) Array is collection of homogenous element
- 2) Array is fixed in size , once the block of memory is created than it cannot be increased or decreased
- 3) It doesn't have any inbuilt method or function to perform to crud operation

Array reference variable :

A variable which is used to store the address of an array object is known as array reference variable.

Index:-

Index is a positive integer number which start from zero(0), it indicates or points this specific position in an array

Syntax of array reference variable :

Datatype [] identifier ;

Example :

Int [] arr;

Syntax to create array object:-

1) using new keyword :-

Syntax:- new Datatype [Size];

Example:- new int[5];

Full syntax:-

Int [] arr=new int [5];

1) Basic programmer of array:

```
package array.org;
public class Array_1 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int [] arr=new int[5];
        arr[0]=10;
        arr[1]=20;
        arr[2]=30;
        arr[3]=40;
        arr[4]=50;
        System.out.println("The length of array is: "+arr.length);
        for(int i=0;i<arr.length;i++)
        {
            System.out.println(arr[i]);
        }
    }
}
```

OUTPUT:-

```
The length of array is: 5
10
20
30
40
50
```

2) Getting the input from user.

```
package array.org;
import java.util.Scanner;
public class Array_2 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element: ");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        for(int i=0;i<arr.length;i++)
        {
            System.out.print(" "+arr[i]);
        }
    }
}
```

OUTPUT:-

```
Enter the size of array:
5
Enter the 5 Element:
10
20
30
```

```
40
50
10 20 30 40 50
```

3)WAP to print even element in the array.

```
package array.org;
import java.util.Scanner;
public class Array_3 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element: ");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("The even elements is: ");
        for(int i=0;i<arr.length;i++)
        {
            if(arr[i]%2==0)
            {
                System.out.println(arr[i]);
            }
        }
    }
}
```

```
OUTPUT:
Enter the size of array:
5
Enter the 5 Element:
10
37
65
14
2
The even elements is:
10
14
2
```

4)WAP to count the element present in array without using length variable.

```
for(int i=1;i<=num;i++)

{

    if(num%i==0)

    {

        count++;
    }
}
```

```
}
```

```
}
```

5)WAP to Identify sum of element present in an array.

```
package array.org;
import java.util.Scanner;
public class Array_4 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int count=0;
        System.out.println("Enter the "+ arr.length + " Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        for(int i=0;i<arr.length;i++)
        {
            count++;
        }

        System.out.println("The count of element is: "+count);
    }
}
```

6)WAP to Identify the Average of Elements in array.

```
package array.org;
import java.util.Scanner;
public class Array_5 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int sum=0;
        System.out.println("Enter the "+ arr.length + " Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.print("The sum of element is: ");
        for(int i=0;i<arr.length;i++)
        {
            sum=sum+arr[i];
        }
    }
}
```

```

        System.out.print(sum);
    }
}

```

7)WAP to Identify the Average of Odd element in array.

```

package array.org;
import java.util.Scanner;
public class Array_7 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int sum=0;
        int Avg;
        int count=0;
        System.out.println("Enter the "+ arr.length +" Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.print("The Avg of element is: ");
        for(int i=0;i<arr.length;i++)
        {
            if(arr[i]%2!=0)
            {
                sum=sum+arr[i];
                count++;
            }
        }
        System.out.print(Avg=sum/count);
    }
}

```

8)WAP to Identify the summation of Even index element.

```

package array.org;

import java.util.Scanner;

public class Array_8 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int sum=0;
        System.out.println("Enter the "+ arr.length +" Element:
");
    }
}

```

```

        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.print("The Avg of element is: ");
        for(int i=0;i<arr.length;i++)
        {
            if(i%2==0)
            {
                sum=sum+arr[i];
            }
        }
        System.out.print(sum);
    }
}

```

9)WAP to Identify the prime Number present in Array.

```

package array.org;

import java.util.Scanner;

public class Array_9 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int sum=0;

        System.out.println("Enter the "+ arr.length +" Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("The prime Number presnt in element is:
");
        for(int i=0;i<arr.length;i++)
        {
            int count=0;
            for(int j=2;j<=arr[i]/2;j++)
            {
                if(arr[i]%j==0)
                {
                    count++;
                }
            }
            if(count==0)
            {

```

```

        System.out.println(arr[i]);
    }
}
}
}

```

10) WJAP To Identify the number of positive or negative element in array.

```

package array.org;
import java.util.Scanner;
public class Array_10 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int pos=0;
        int neg=0;
        System.out.println("Enter the "+ arr.length + " Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        for(int i=0;i<arr.length;i++)
        {
            if(arr[i]>0)
            {
                pos++;
            }
            else
            {
                neg++;
            }
        }
        System.out.println("The positive Ele are: "+pos);
        System.out.println("The negative Ele are: "+neg);
    }
}

```

11) WJAP the summation of each factorial of element present in array.

```

package array.org;

import java.util.Scanner;

public class Array_11 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
    }
}

```

```

    int []arr=new int[size];
    int sum=0;
    System.out.println("Enter the "+ arr.length +" Element:
");
    for(int i=0;i<arr.length;i++)
    {
        arr[i]=sc.nextInt();
    }
    System.out.print("The sum of element is: ");
    for(int i=0;i<arr.length;i++)
    {
        int fact=1;
        for(int j=1;j<=arr[i];j++)
        {
            fact=fact*j;
        }
        sum=sum+fact;
    }
    System.out.println(sum);
}
}

```

12)WAP to reverse the element present in array.

```

package array.org;
import java.util.Scanner;
public class Array_12 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int start=0;
        int end=arr.length-1;
        System.out.println("Enter the "+ arr.length +" Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        //logic the reverse the array
        while(start<end)
        {
            int temp=arr[start];
            arr[start]=arr[end];
            arr[end]=temp;
            start++;
            end--;
        }
        for(int i=0;i<arr.length;i++)
        {

```



```

        System.out.print(" "+arr[i]);
    }
}

```

13)WAP for array palindrome.

```

package array.org;
import java.util.Scanner;
public class Array_13 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        //logic to Reverse array & store in another array object
        int []arr1=new int[arr.length];
        int j=0;
        int count=0;
        for(int i=arr.length-1;i>=0;i--)
        {
            arr1[j]=arr[i];
            j++;
        }
        //logic for palindrom
        for(int i=0;i<arr.length;i++)
        {
            if(arr[i]==arr1[i])
            {
                count++;
            }
            else break;
        }
        System.out.println("the count is: "+count);
        if(arr.length==count)
        {
            System.out.println("palindrom");
        }
        else
        {
            System.out.println("NOT palindrom");
        }
    }
}

```

14)WAJA for array palindrome another way.

```
package array.org;
import java.util.Scanner;
public class Array_14 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int start=0;
        int end=arr.length-1;
        int count=0;
        while(start<end)
        {
            if(arr[start]==arr[end])
            {
                count++;
            }
            else break;
            start++;
            end--;
        }
        if(count==arr.length/2)
        {
            System.out.println("palindrom");
        }
        else
        {
            System.out.println("NOT palindrom");
        }
    }
}
```

15)WAJP to Identify the smallest element in array.

```
package array.org;
import java.util.Scanner;
public class Array_15 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
```

```

        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int small=arr[0];
        for(int i=1;i<arr.length;i++)
        {
            if(arr[i]<small)
            {
                small=arr[i];
            }
        }
        System.out.println("The smallest element is: "+small);
    }
}

```

16)WAP to Identify the largest element in array.

```

package array.org;
import java.util.Scanner;
public class Array_16 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int large=arr[0];
        for(int i=1;i<arr.length;i++)
        {
            if(arr[i]>large)
            {
                large=arr[i];
            }
        }
        System.out.println("The smallest element is: "+large);
    }
}

```

17)WAP to Identify the second smallest element in array.

```

package array.org;
import java.util.Scanner;
public class Array_17 {

```

```

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int small=arr[0];
        int small2=Integer.MAX_VALUE;
        for(int i=1;i<arr.length;i++)
        {
            if(arr[i]<small)
            {
                small2=small;
                small=arr[i];
            }
            else if(small2>arr[i] && small !=arr[i])
            {
                small2=arr[i];
            }
        }

        System.out.println("The 1st smallest element is: " +
small);
        System.out.println("The 2nd smallest element is: " +
small2);

    }
}

```

18)WJJP to Identify the second largest element in array.

```

package array.org;
import java.util.Scanner;
public class Array_18 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int large=arr[0];

```

```

int large2=Integer.MIN_VALUE;
for(int i=1;i<arr.length;i++)
{
    if(arr[i]>large)
    {
        large2=large;
        large=arr[i];
    }
    else if(large2<arr[i] && large !=arr[i])
    {
        large2=arr[i];
    }
}

System.out.println("The 1st largest element is: " +
large);
System.out.println("The 2nd largest element is: " +
large2);

}
}

```

19)WAP to search an element in array.

```

package array.org;
import java.util.Scanner;
public class Array_19 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        int sum=0;
        System.out.println("Enter the "+ arr.length +" Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.print("Enter the Number which is search: ");
        int Ele=sc.nextInt();
        int count=0;
        for(int i=0;i<arr.length;i++)
        {
            if(Ele==arr[i])
            {
                count++;
                break;
            }
        }
        if(count>0)
        {

```

```

        System.out.println("Element found !!!!");
    }
    else
    {
        System.out.println("Element Not found !!!!");
    }
}
}

```

20)WJJP to the highest sum of two number.

```

package array.org;
import java.util.Scanner;
public class Array_20 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        int large=arr[0];
        int large2=Integer.MIN_VALUE;
        for(int i=1;i<arr.length;i++)
        {
            if(arr[i]>large)
            {
                large2=large;
                large=arr[i];
            }
            else if(large2<arr[i] && large !=arr[i])
            {
                large2=arr[i];
            }
        }

        System.out.println("The 1st largest element is: " +
large);
        System.out.println("The 2nd largest element is: " +
large2);
        int sum=large+large2;
        System.out.println("The sum of Highest two number is:
"+sum);
    }
}

```

21)WAJP to the highest sum of two number. Another way !!!!

```
package array.org;
import java.util.Scanner;
public class Array_21 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        //logic to identify the Highestsum
        int sum=0;
        for(int i=0;i<arr.length;i++)
        {
            for(int j=i+1;j<arr.length;j++)
            {
                if(sum<arr[i]+arr[j])
                {
                    sum=arr[i]+arr[j];
                }
            }
        }
        System.out.println("The highest sum of two number: "+sum);
    }
}
```

22)WAJP TO Identify the pair of elements is equals to given Number.

```
package array.org;
import java.util.Scanner;
public class Array_22 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("Enter the Ele of which you want to find pair: ");
        int num=sc.nextInt();
```

```

        //logic to identify the Pair
        int sum=0;
        int count=0;
        for(int i=0;i<arr.length;i++)
        {
            for(int j=i+1;j<arr.length;j++)
            {
                if(num==arr[i]+arr[j])
                {
                    System.out.println("Pair are: "+arr[i]+" &
"+arr[j]);
                    count++;
                }
            }
        }
        if(count==0)
        {
            System.out.println("No Pair are found");
        }
    }
}

```

23)WAP to reverse the each element in array.

```

package array.org;

import java.util.Arrays;
import java.util.Scanner;

public class Array_23 {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the size of array:");

        int size=sc.nextInt();

        int []arr=new int[size];

        System.out.println("Enter the "+ arr.length +" Element: ");

        for(int i=0;i<arr.length;i++)

        {

            arr[i]=sc.nextInt();

```



```

    }

    int arr1[]=new int [arr.length];

    for(int i=0;i<arr.length;i++)

    {

        int rev=0;

        int num=arr[i];

        while(num!=0)

        {

            int ld=num%10;

            rev=rev*10+ld;

            num=num/10;

        }

        arr1[i]=rev;

    }

    System.out.println(Arrays.toString(arr));

    System.out.println(Arrays.toString(arr1));

```

```

    }

```

```

}

```

24)WAJP to reverse an array also reverse the each element.

```

package array.org;

```

```

import java.util.Arrays;

```

```

import java.util.Scanner;

```

```
public class Array_24 {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc=new Scanner(System.in);  
        System.out.println("Enter the size of array:");  
        int size=sc.nextInt();  
        int []arr=new int[size];  
        System.out.println("Enter the "+ arr.length +" Element: ");  
        for(int i=0;i<arr.length;i++)  
        {  
            arr[i]=sc.nextInt();  
        }  
        int arr1[]=new int [arr.length];  
        int k=arr.length-1;  
        for(int i=0;i<arr.length;i++)  
        {  
            int rev=0;  
            int num=arr[i];  
            while(num!=0)  
            {  
                int ld=num%10;  
                rev=rev*10+ld;  
                num=num/10;  
            }  
            arr1[k]=rev;  
            k--;  
        }  
    }  
}
```

```

    }

    System.out.println(Arrays.toString(arr));

    System.out.println(Arrays.toString(arr1));

}

```

25)WAP to remove duplicate element form an array.

```

package array.org;
import java.util.Scanner;
public class Array_25 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:
");

        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("Remove the duplicates : ");
        for(int i=0;i<arr.length;i++)
        {
            if(arr[i]!=-1)
            {
                for(int j=i+1;j<arr.length;j++)
                {
                    if(arr[i]==arr[j])
                    {
                        arr[j]=-1;
                    }
                }
            }
            if(arr[i]!=-1)
            {
                System.out.println(arr[i]);
            }
        }

    }
}

```

26)WAP to identify the duplicates element in array.

```
package array.org;

import java.util.Scanner;

public class Array_26 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:");

        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("Identify the duplicates : ");
        for(int i=0;i<arr.length;i++)
        {

            for(int j=i+1;j<arr.length;j++)
            {
                if(arr[i]==arr[j])
                {
                    System.out.println(arr[i]);
                }
            }
        }
    }
}
```

27)WAP to find out the frequency of each element.

```
package array.org;
import java.util.Scanner;
public class Array_27 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length +" Element:");

        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("Identify the duplicates : ");
        for(int i=0;i<arr.length;i++)
```

```

        {
            int count=1;
            if(arr[i]!=-1)
            {
                for(int j=i+1;j<arr.length;j++)
                {
                    if(arr[i]==arr[j])
                    {
                        count++;
                        arr[j]=-1;
                    }
                }
                System.out.println(arr[i]+" is present "+ count + "
Time");
            }
        }
    }
}

```

28)WAP to print the frequency of duplicate element

```

package array.org;

import java.util.Scanner;

public class Array_28 {
    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the size of array:");
        int size=sc.nextInt();
        int []arr=new int[size];
        System.out.println("Enter the "+ arr.length + " Element:
");
        for(int i=0;i<arr.length;i++)
        {
            arr[i]=sc.nextInt();
        }
        System.out.println("Identify the duplicates : ");
        for(int i=0;i<arr.length;i++)
        {
            int count=1;
            if(arr[i]!=-1)
            {
                for(int j=i+1;j<arr.length;j++)
                {
                    if(arr[i]==arr[j])
                    {
                        count++;
                        arr[j]=-1;
                    }
                }
            }
        }
    }
}

```

```

        }
        if(count>1)
        {
            System.out.println(arr[i]+" is present "+ count +"
Time");
        }
    }
}
}
}

```

29)WAJP to marge to arrays.

```

package array.org;

import java.util.Arrays;

import java.util.Scanner;

public class Array_29 {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the 1st size of array:");

        int size=sc.nextInt();

        int []arr=new int[size];

        System.out.println("Enter the "+ arr.length +" Element: ");

        for(int i=0;i<arr.length;i++)

        {

            arr[i]=sc.nextInt();

        }

        System.out.println("Enter the 2nd size of array:");

        int size1=sc.nextInt();

        int []arr1=new int [size1];
    }
}

```

```
System.out.println("Enter the "+ arr1.length +" Element: ");

for(int i=0;i<arr1.length;i++)

{

    arr1[i]=sc.nextInt();

}

System.out.println("Array 1 & 2 :==> ");

System.out.println(Arrays.toString(arr));

System.out.println(Arrays.toString(arr1));


//logic of merging array

int []arr3=new int[arr.length+arr1.length];

int j=0;


for(int i=0;i<arr3.length;i++)

{

    if(i<arr.length)

    {

        arr3[i]=arr[i];

    }

    else

    {

        arr3[i]=arr1[j];

        j++;

    }

}

System.out.println(Arrays.toString(arr3));
```

```

    }

}
30)WBJP to insert in an element in particular position or index of the
array(manes create new array).

package array.org;

import java.util.Arrays;

import java.util.Scanner;

public class Array_30 {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the size of array:");

        int size=sc.nextInt();

        int []arr=new int[size];

        System.out.println("Enter the "+ arr.length +" Element: ");

        for(int i=0;i<arr.length;i++)

        {

            arr[i]=sc.nextInt();

        }

        //logic for insert element;

        int []mod=new int [arr.length+1];

        System.out.println("Enter the index postion: ");

        int pos=sc.nextInt();

        System.out.println("Enter the element to be insatrted in given
postion : ");

        int ele=sc.nextInt();

```



```

        if(pos<0 || pos>size-1)
        {
            System.out.println("invalid position");
        }
        else
        {
            for(int i=arr.length-1;i>=pos;i--)
            {
                mod[i+1]=arr[i];
            }
            mod[pos]=ele;
            for(int i=0;i<pos;i++)
            {
                mod[i]=arr[i];
            }
            System.out.println(Arrays.toString(arr));
            System.out.println(Arrays.toString(mod));
        }
    }
}

```

31)WAJP to insert in element in particular position or index of array in same array(manes don't create new array modify same array)

```
package array.org;
```

```
import java.util.Arrays;
```

```
import java.util.Scanner;
```

```
public class Array_31 {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc=new Scanner(System.in);  
        System.out.println("Enter the size of array:");  
        int size=sc.nextInt();  
        int []arr=new int[size+1];  
  
        System.out.println("Enter the "+ size +" Element: ");  
        for(int i=0;i<size;i++)  
        {  
            arr[i]=sc.nextInt();  
        }  
        //logic to insert element  
        System.out.println("Enter the index postion: ");  
        int pos=sc.nextInt();  
        System.out.println("Enter the element to be insatrtrted in given  
postion : ");  
        int ele=sc.nextInt();  
        if(pos<0 || pos>size)  
        {  
            System.out.println("invalid position");  
        }  
        else  
        {  
            for(int i=arr.length-1;i>pos;i--)
```

```

        {
            arr[i]=arr[i-1];
        }

        arr[pos]=ele;

        System.out.println(Arrays.toString(arr));
    }

```

```

    }

```

```

}

```

32)WAP to remove the element in specific position or index of the array(use same array).

```

package array.org;

```

```

import java.util.Arrays;

```

```

import java.util.Scanner;

```

```

public class Array_32 {

```

```

    public static void main(String[] args) {

```

```

        // TODO Auto-generated method stub

```

```

        Scanner sc=new Scanner(System.in);

```

```

        System.out.println("Enter the size of array:");

```

```

        int size=sc.nextInt();

```

```

        int []arr=new int[size];

```

```

        System.out.println("Enter the "+ arr.length +" Element: ");

```

```

        for(int i=0;i<arr.length;i++)

```

```

        {

```

```

        arr[i]=sc.nextInt();
    }

    //logic to remove element

    System.out.println("Enter the index postion: ");

    int pos=sc.nextInt();

    for(int i=pos;i<arr.length-1;i++)
    {
        arr[i]=arr[i+1];
    }

    //acces the element

    System.out.println("The result is: ");

    for(int i=0;i<arr.length-1;i++)
    {
        System.out.println(arr[i]);
    }

}

```

33)WAJP to remove the element in specific position or index of the array.

```
package array.org;
```

```
import java.util.Arrays;
```

```
import java.util.Scanner;
```

```
public class Array_33 {
```

```
    public static void main(String[] args) {
```

```
        // TODO Auto-generated method stub
    }
}

```

```
Scanner sc=new Scanner(System.in);

System.out.println("Enter the size of array:");

int size=sc.nextInt();
int []arr=new int[size];


System.out.println("Enter the "+ arr.length +" Element: ");
for(int i=0;i<arr.length;i++)
{
    arr[i]=sc.nextInt();
}

//logic to remove element

System.out.println("Enter the index postion: ");
int pos=sc.nextInt();
int []b=new int[arr.length-1];
int j=0;
for(int i=0;i<arr.length;i++)
{
    if(i==pos)
    {
        i++;
    }
    b[j++]=arr[i];
}

System.out.println(Arrays.toString(arr));

System.out.println(Arrays.toString(b));

}
```

```
}

34WAJP to marge zigzag way arrays.

package array.org;

import java.util.Arrays;

import java.util.Scanner;

public class Array_34 {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the 1st size of array:");

        int size=sc.nextInt();

        int []arr=new int[size];

        System.out.println("Enter the "+ arr.length +" Element: ");

        for(int i=0;i<arr.length;i++)

        {

            arr[i]=sc.nextInt();

        }

        System.out.println("Enter the 2nd size of array:");

        int size1=sc.nextInt();

        int []arr1=new int [size1];

        System.out.println("Enter the "+ arr1.length +" Element: ");

        for(int i=0;i<arr1.length;i++)

        {

            arr1[i]=sc.nextInt();

        }

    }

}
```

```

System.out.println("Array 1 & 2 :==> ");

System.out.println(Arrays.toString(arr));

System.out.println(Arrays.toString(arr1));


//creating array object arr1+arr2

int []arr3=new int[arr.length+arr1.length];

int x=0;

int y=0;

int z=0;

while(z<arr3.length)

{

    if(x<arr.length)

    {

        arr3[z++]=arr[x++];

    }

    if(y<arr1.length)

    {

        arr3[z++]=arr1[y++];

    }

}

System.out.println(Arrays.toString(arr3));

}

```

35)WAFP to swap the first half of the array element to the second half.

```

package array.org;

import java.util.Arrays;

```

```
import java.util.Scanner;

public class Array_35 {

    public static void main(String[] args) {

        // TODO Auto-generated method stub

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter the size of array:");

        int size=sc.nextInt();

        int []arr=new int[size];

        System.out.println("Enter the "+ arr.length +" Element: ");

        for(int i=0;i<arr.length;i++)

        {

            arr[i]=sc.nextInt();

        }

        System.out.println("Before swapping: "+Arrays.toString(arr));

        System.out.println("Swap the first half of the array element to the second half ");

        int mid=0;

        if(arr.length%2==0)

        {

            mid=arr.length/2;

        }

        else

        {

            mid=arr.length/2+1;

        }

        for(int i=0;i<arr.length/2;i++)

        {
```



```
        int temp=arr[i];

        arr[i]=arr[mid];

        arr[mid]=temp;

        mid++;

    }

    System.out.println("After swapping: "+Arrays.toString(arr));

}

}
```